proprietary data feeds, or a newly developed consolidated market data feed.[795] However, if an SRO developed a dedicated consolidated market data feed, the SRO will have to take steps to ensure that any proprietary data feed is not made available on a more timely basis (i.e., by any time increment that could be measured by the SRO) than a consolidated market data feed.[796] Rule 603(a) also applies to the provision of data content underlying consolidated market data by the SROs to competing consolidators. The Commission believes that under Rule 603(a), if an SRO developed a consolidated market data feed, it would likely have to throttle any order-by-order proprietary data feed so that it is not made available on a more timely basis than such dedicated consolidated market data feed. Any dedicated consolidated market data feed developed by an exchange would likely involve processing by an exchange to segment its consolidated market data elements, which adds latency to data dissemination, while an order-by-order proprietary data feed would involve no less processing by the exchange. Further, the SROs are allowed to offer different access options (e.g., with different latencies, throughput capacities, and data-feed protocols) to market data customers, as long as any access options available to proprietary data customers are made available to competing consolidators and self-aggregators for their selection for the collection of the data necessary to generate consolidated market data.

---

[795] Competing consolidators and self-aggregators would be permitted to choose among the data feed options offered by the SROs to satisfy their obligations under Rule 603(b) to collect the SRO information that is necessary to generate consolidated market data. See also Section III.E.2(g) for a discussion of the licensing, billing and audit process. The effective national market systems plans through their licensing, billing and audit processes can determine the extent to which data content utilized by competing consolidators and self-aggregators constituted elements of consolidated market data.

[796] See infra Section III.B.9(f).

In addition, if an SRO provided its proprietary data products to competing consolidators or self-aggregators for purposes of Rule 603(b) and a competing consolidator or self-aggregator developed a product, or otherwise used data content that is beyond the scope of consolidated market data (e.g., full depth of book data), the proprietary data content will be subject to the individual exchange fees for proprietary data,[797] while the data content underlying consolidated market data will be subject to the fees established pursuant to the effective national market system plan(s).[798]

The Commission received several comments on its proposed amendments to Rule 603(b). Some commenters stated that the changes to Rule 603(b) were key to the success of the proposal.[799] One commenter stated that the proposed changes to Rule 603(b) "should help to ensure that proprietary feeds, competing consolidators and self-aggregators operate on a more level playing field with regards to the speed that market participants can obtain market data and access."[800] The commenter continued that the proposal would create a true alternative to subscribing to and paying for each individual exchange's proprietary feed.[801] Another

---

797 Fees for proprietary data are filed with the Commission pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder.

798 Fees for the data content underlying consolidated market data will be filed with the Commission pursuant to Rule 608 of Regulation NMS, 17 CFR 242.608.

799 See McKay Letter; SIFMA Letter. See also Clearpool Letter at 7 (supporting the proposed language of Rule 603(b)).

800 SIFMA Letter at 11. See also McKay Letter at 4 (stating that the proposed language of Rule 603(b) would prevent an exchange from creating a proprietary data feed that would not be sufficient to create consolidated market data but which has a latency or other access advantage associated with it).

801 See SIFMA Letter at 11.

commenter stated that the requirement to make data available to competing consolidators in the same manner as the exchanges make it available to any other person would address latency issues that exist in core data.[802]

One commenter suggested that the exchanges provide content that is similar to what they provide on their direct feeds, stating that all data can be useful for "execution algorithms' quantitative trading decisions."[803] Another commenter suggested that the exchanges should provide a single data feed—their proprietary DOB feed—to satisfy the requirements of the proposed rules, simplify data distribution, and ease Rule 603(a) burdens.[804] The Commission has set forth minimum data content underlying consolidated market data that must be made available by the SROs under Rule 603(b). The Commission has identified these elements as necessary for a wide array of trading in the national market system. Because the Commission recognizes that some market participants may need more information than what is defined as consolidated market data, SROs can provide additional information to these customers via proprietary feeds. Further, as discussed above, the SROs can satisfy their obligations under Rule 603(b) by utilizing their proprietary data feeds, a combination of proprietary data feeds, or a newly developed core data feed that contains all of the data content underlying consolidated market data.[805] Rule 603(b) does not specify a required method of delivery of the data content underlying consolidated market data but requires that such data be provided in the same manner and using

---

802 See Fidelity Letter at 10.

803 BestEx Research Letter at 5.

804 See MEMX Letter at 9.

805 See supra note 795 and accompanying text.

the same methods of access and the same format,[806] as the SROs use to make any NMS information available in NMS stocks to any person.

One commenter, however, stated, without further explanation, that the "same manner and methods" language was "merely a standard price list offered by the exchange."[807] The commenter further stated that the "same format" language would hurt average investors and give high frequency trading firms an advantage and asserted that "[o]ne can only attempt to match faster connectivity by altering data format and compression methods."[808] To the extent that the commenter is suggesting that investors who obtain consolidated market data products from a competing consolidator may be at a disadvantage to self-aggregators because of the latency advantage of self-aggregators, the Commission believes, as discussed in Section IV.D.2(d), that the self-aggregators will not necessarily have a systematic latency advantage over competing consolidators. If the commenter is suggesting that SROs provide data in a different format to compensate for faster connectivity, Rule 603(b) requires the SROs to make the data content underlying consolidated market data available to competing consolidators and self-aggregators in the same manner and methods, including all methods of access and the same format, as proprietary data. Further, Rule 603(a) prohibits an SRO from making NMS information available to any person on a more timely basis (i.e., by any time increment that could be

806 See Proposing Release, 85 FR at 16770.

807 Data Boiler Letter II at 1.

808 Id. at 1.

measured by the SRO) than it makes such data available to competing consolidators and self-aggregators.[809]

Another commenter stated that the proposed language in Rule 603(b) did not consider that an exchange's connectivity options may not have the capacity to be provided to all competing consolidators and self-aggregators in the same manner and using the same methods.[810] The commenter stated that the proposal did not address the possible impact on wireless connectivity or how customers would be affected if the SROs ceased to offer wireless connectivity. The commenter did not expand upon how the decentralized consolidation model would burden capacity beyond what occurs today in the provision of proprietary data using the same connectivity options. The exchanges assess capacity needs for their proprietary data products and will be able to assess such needs for the connectivity options for the data content underlying consolidated market data. Each individual SRO will be required to provide the data underlying consolidated market data to competing consolidators and self-aggregators in the same manner and using the same methods as is provided for proprietary data. The exchanges can utilize their proprietary data feeds to make the data content underlying consolidated market data available to competing consolidators and self-aggregators and will be able to offer any access options available to proprietary data feeds to competing consolidators and self-aggregators for their selection.

(c) Comments on Access Options

In the Proposing Release, the Commission stated that the exchanges could provide different access options (e.g., with different latencies, throughput capacities, and data feed

---

809 See infra Section III.B.9(f).

810 See IDS Letter I at 11.

protocols) to market data customers, but any access options available to proprietary data customers must also be available to competing consolidators and self-aggregators. As proposed, Rule 603(b) would require exchanges to provide all forms of access used for proprietary data to all competing consolidators and self-aggregators for the collection of the data necessary to generate proposed consolidated market data.[811] Further, an exchange must offer the same form of access, such as fiber optics, wireless, or other forms, in the same manner and using the same methods, including all methods of access and the same format, as the exchange offers for its proprietary data. For instance, if an exchange has more than one form of transmission for its proprietary data, then the exchange must offer the competing consolidators and self-aggregators those types of transmission for consolidated market data. As discussed, the rule will not require an exchange to offer new forms of access, but if an exchange did offer any new forms of access for proprietary data, it will have to offer them for consolidated market data as well.

One commenter stated that all forms of data submission must assure full and equal, transparent, and equally timed access to data.[812] Another commenter stated that the most important component for the success of the competing consolidator model is "to ensure that all market participants have the opportunity for equal access within the facilities of an exchange for purposes of receiving market data from the exchange, transmitting that market data out of the exchange and receiving and distributing market data from another exchange (e.g., to receive and deliver an away exchange's market data."[813] This commenter further stated that exchanges

---

811 See Proposing Release, 85 FR at 16770.

812 See Kubitz Letter.

813 McKay Letter at 3. See also Temple University Letter at 2 (stating that "intermediaries" including exchanges and SIPs should not "unfairly discriminate or privilege certain

should not favor certain market participants over others and that equal access must cover both egress and ingress within exchange data centers.[814] Further, the commenter stated that the economic incentive to become a competing consolidator may not be sufficient without such equal access.[815] Another commenter cautioned that allowing an exchange to develop a new market data product that contains only the data elements that are specified in the proposed definition of consolidated market data could be problematic because (1) competing consolidators likely will have feed handlers for publicly available feeds, which would impose additional burdens; and (2) there would be no incentive for exchanges to keep the latency of these special feeds similar to that of public feeds.[816]

The Commission believes that the SROs should be required to provide the information necessary to generate consolidated market data in the same manner and using the same methods, including all methods of access and the same format as they provide to any person. Different forms of access affect the delivery of data. If an exchange provided a superior form of access to its proprietary data products, then transmission of consolidated market data would be negatively impacted and the benefits of the decentralized consolidation model, such as lower latencies, would not be realized. The national market system would be affected by this disparity as it is today—with certain proprietary data products providing superior access as compared to consolidated market data products. As stated in the Proposing Release, different market

---

market participants to the detriment of others in terms of data access, execution and other market data operations" and that such intermediaries should act "independently and neutrally towards market participants to ensure the competitive integrity of the marketplace").

814 See McKay Letter at 3.

815 Id.

816 See NovaSparks Letter at 1–2.

participants have different access needs. The Commission is not mandating a specific access option or limiting options for market participants, but all access options, including co-location, must be available to all market participants whether they are purchasing the data content underlying consolidated market data or proprietary data.

The access requirement under Rule 603(b) requires the exchanges to provide their NMS information, including all data necessary to generate consolidated market data, at one data dissemination location co-located near each exchange's matching engine. The requirement in Rule 603(b) that access be provided in the same manner as any other information is provided to any person encompasses co-location options that are provided by exchanges to market participants that purchase proprietary data. Proprietary data users are typically co-located near an exchange's matching engine. Rule 603(b) will allow competing consolidators and self-aggregators to receive data at the same speeds, and with the same access options, as the exchange offers its market data.[817] Different co-location options within a data center could raise concerns about whether that exchange is providing the same manner of access to its data as required under Rule 603(b). Further, the exchanges would not be permitted to provide their NMS information necessary to generate consolidated market data in a faster manner to any affiliate exchange, a subsidiary, or other affiliate that operates as a competing consolidator or a subsidiary or affiliate that competes in the provision of proprietary data.

(d) Comments on Latency Neutralization

In the Proposing Release, the Commission stated that proposed Rule 603(b) would require that all access options be provided in a latency-neutralized manner such that all participants within an exchange's data center—such as proprietary data subscribers, competing

---

[817] See also infra Section III.B.9(f).

consolidators, and self-aggregators—would receive information at the same time, regardless of their location or status within the data center. The Commission continued that exchanges could, for example, adopt equal cable length protocols (i.e., where cable lengths from network equipment to customer cabinets are harmonized for equal access) to ensure that all of the exchange's data center connections provide market data simultaneously. Finally, the Commission stated that the SROs must use the same latency-neutralization processes for competing consolidators and self-aggregators as they offer subscribers of proprietary data.

One commenter stated that to provide a level playing field "latency-neutralized" access must apply to locally produced data (i.e., data produced within an exchange's data center where a market participant is co-located) and that the exchange must not interfere in the competition to provide inbound market data from exchanges located in other market centers.[818] This commenter provided a detailed discussion of five steps of data access and delivery that it believed should be subject to latency neutralization requirements under Rule 603(b) if an exchange or its affiliate exercises direct control or indirect control.[819] Specifically, the commenter stated that latency neutralization should apply to (1) the initial distribution from an exchange's market data distribution engine to the cabinets of a competing consolidator, self-aggregator, or other direct recipient of market data; (2) the distribution from a competing consolidator's cabinet out of an

---

818 See McKay Letter at 6.

819 See id. at 7. The commenter suggested that "indirect control" means the use of the exchange or its affiliate's influence, weight or pressure to create an advantage or disadvantage in exchange connectivity to select market participants. Further, the commenter described several different forms of direct or indirect control, including "requiring market participants to connect to a meet me room, specifying the types of cross connects that may be used, restricting the use of certain frequencies to certain market participants, through the use of one or more affiliates or select third parties to create advantages, or pursuant to formal and informal arrangements with the data center operator." Id. at 8–9.

exchange's data center to equipment (e.g., wireless or fiber equipment) for distribution to subscribers (described as the "egress leg"); (3) the receipt of market data from an away exchange for delivery to co-located customers in the exchange's data center (described as the "ingress leg"); (4) the delivery of data from a competing consolidator's cabinet of market data received from away markets to the competing consolidator's subscribers that are located in the same data center (described as the "delivery to subscriber leg"); and (5) the transmission of data from an exchange data center to be received at other exchange data centers and/or for distribution to non-co-located market participants (described as the "transit leg"). This commenter stated that these considerations are important to determining when and where competition begins in providing consolidated market data and that it believes that competition should begin when market data leaves those areas over which an exchange or its affiliates exercise direct or indirect control.[820]

Rule 603(b) requires that the exchanges provide their NMS information, including all data necessary to generate consolidated market data, in the same manner and using the same methods as such exchange provides any information to any person. As discussed above, this language encompasses the provision of data by an exchange at one data dissemination location co-located near each exchange's matching engine to allow competing consolidators and self-aggregators to receive data at that location at the same speeds, and with the same access options, as the exchange offers its market data. The SROs are required to use the same latency-neutralization processes for competing consolidators and self-aggregators as they offer to subscribers of proprietary data such that all participants within the exchange's data center,

---

820 See id. at 10.